

Overhead camera · one ROI per approach

ROI 0

ROI 1

ROI 2

ROI 3

Per-approach estimator $\hat{\phi}$ (YOLOv11 + ByteTrack)

each ROI $\rightarrow$ 5 bounded features:

queue · occupancy · speed · moto-share · heavy-share

20 approach features (4×5)

queue values only

Signed pressure · 1 DERIVED dim
 $\text{clip}(0.5 \cdot (1 + \bar{q}_A - \bar{q}_B), 0, 1)$
reuses only the 4 queue values
(groups A / B)

Controller-internal
(not camera):
phase one-hot (4)
+ green-timer (1)

Local observation $\mathbf{O}_i$ — 26 dims
20 approach + 4 phase + 1 timer + 1 pressure